

Geometric Properties in Writing Proofs

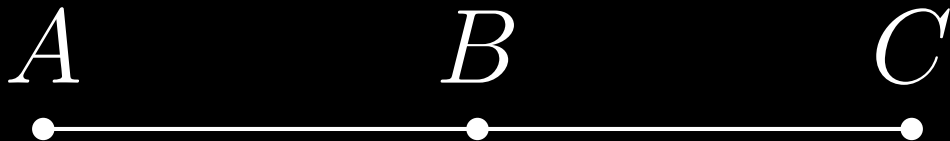
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Definitions

Betweenness

If B is between A and C on $\overline{AC}$, then

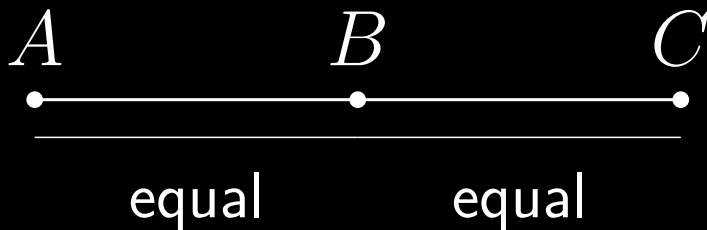
$$AC = AB + BC$$



Midpoint

If B is the midpoint of $\overline{AC}$, then

$$AB = BC$$



Segment Bisector

If a line, ray, or segment bisects $\overline{AB}$ at X , then

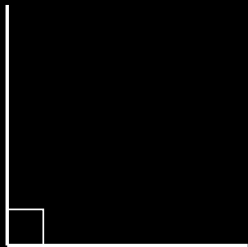
$$AX \cong BX$$

A bisector always cuts a segment into two congruent parts at its midpoint X .

Right Angle

If $\angle A$ is a right angle, then

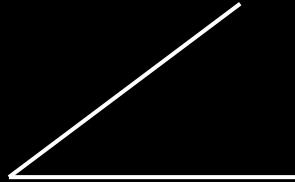
$$m\angle A = 90^\circ$$



Acute Angle

If $\angle A$ is an acute angle, then

$$m\angle A < 90^\circ$$



Obtuse Angle

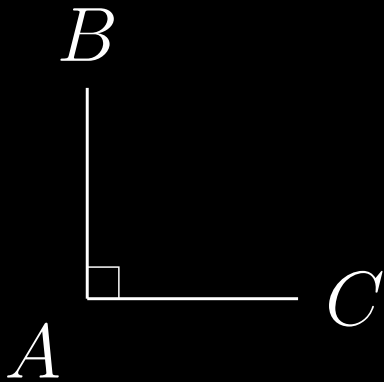
If $\angle A$ is an obtuse angle, then

$$m\angle A > 90^\circ$$



Perpendicular Line Segments

If $\overline{AB} \perp \overline{AC}$, then $\angle BAC$ is a right angle.



Complementary Angles

If $\angle A$ and $\angle B$ are complementary angles, then

$$m\angle A + m\angle B = 90^\circ$$

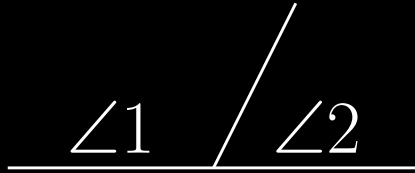
Supplementary Angles

If $\angle A$ and $\angle B$ are supplementary angles, then

$$m\angle A + m\angle B = 180^\circ$$

Linear Pair

If two angles are adjacent such that two of the rays are opposite, then they form a linear pair.



A linear pair is always supplementary.

Angle Bisector

If $\overrightarrow{AD}$ bisects $\angle BAC$, then

$$\angle BAD \cong \angle DAC$$

Congruent Segments

If $\overline{AB} \cong \overline{CD}$, then

$$AB = CD$$

Congruent Angles

If $\angle A \cong \angle B$, then

$$m\angle A = m\angle B$$

Properties of Equality

Addition Property of Equality

If $a = b$, then

$$a + c = b + c$$

Subtraction Property of Equality

If $a = b$, then

$$a - c = b - c$$

Multiplication Property of Equality

If $a = b$, then

$$ac = bc$$

Division Property of Equality

If $a = b$ and $c \neq 0$, then

$$\frac{a}{c} = \frac{b}{c}$$

Reflexive Property of Equality

If a is any real number, then

$$a = a$$

Symmetric Property

If $a = b$, then

$$b = a$$

Transitive Property

If $a = b$ and $b = c$, then

$$a = c$$

Law of Substitution

If $a + b = c$ and $b = x$, then

$$a + x = c$$

Postulates

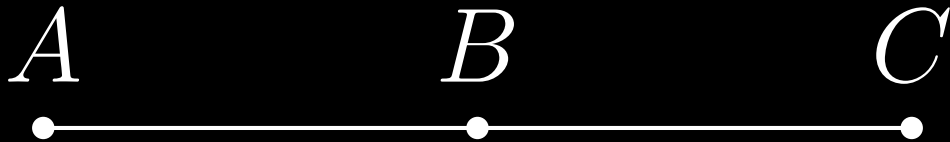
Supplement Postulate

If two angles form a linear pair, then they are supplementary.

Segment Addition Postulate

If B lies on $\overline{AC}$, then

$$AC = AB + BC$$



Angle Addition Postulate

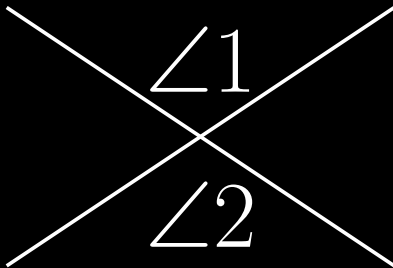
If B is in the interior of $\angle AOC$, then

$$m\angle AOC = m\angle AOB + m\angle BOC$$

Theorems

Vertical Angle Theorem

If two angles are vertical, then they are congruent.



Complement Theorem

If two angles are complements of the same (or congruent) angles, then they are congruent.

Supplement Theorem

If two angles are supplements of the same (or congruent) angles, then they are congruent.

Geometric Properties

Supplement Postulate (SP)

If two angles form a linear pair, then they are supplementary.

Vertical Angle Theorem (VAT)

The measures of vertical angles are equal, or vertical angles are congruent.

Angle Sum of a Point Postulate (ASPP)

The sum of the measures of the angles at a point is

$$360^{\circ}$$

Supplement Theorem (ST)

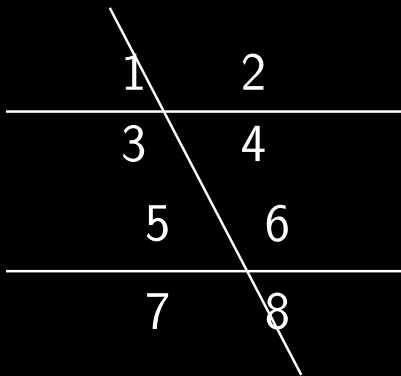
Supplements of congruent angles are congruent.

Complement Theorem (CP)

Complements of congruent angles are congruent.

Parallel Lines & Transversals

Two parallel lines cut by a transversal form eight angles.



Corresponding Angles Postulate (PCAC)

If two parallel lines are cut by a transversal, then corresponding angles are congruent.

Corresponding angles sit in the same position at each intersection, e.g. $\angle 1 \cong \angle 5$.

Alternate Interior Angles Theorem (PAIC)

If two parallel lines are cut by a transversal, then alternate interior angles are congruent.

They lie between the parallel lines, on opposite sides of the transversal, e.g. $\angle 4 \cong \angle 6$.

Alternate Exterior Angles Theorem (PAEC)

If two parallel lines are cut by a transversal, then alternate exterior angles are congruent.

They lie outside the parallel lines, on opposite sides of the transversal, e.g. $\angle 1 \cong \angle 8$.

Same-Side Interior Angles Theorem (PSSIAS)

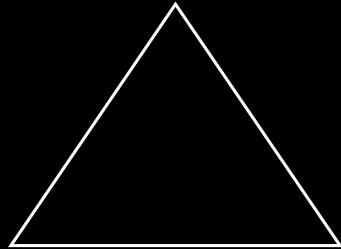
If two parallel lines are cut by a transversal, then same-side interior angles are supplementary.

They lie between the parallel lines, on the same side of the transversal, e.g. $m\angle 4 + m\angle 5 = 180^\circ$.

Triangle Interior Angle Theorem (TIAT)

The sum of the degree measures of the angles
of a triangle is

$$180^{\circ}$$

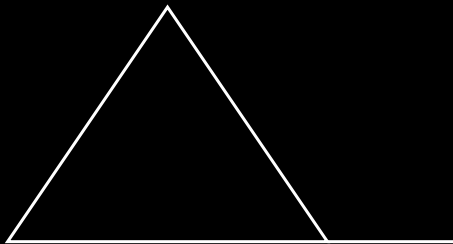


Third Angles Theorem

If two angles of one triangle are congruent to two angles of another triangle, then the third angles are also congruent.

Exterior Angles Theorem (EAT)

The measure of an exterior angle of a triangle equals the sum of the measures of its two remote interior angles.



Quadrilateral Interior Angle Theorem (QIAT)

The sum of the measures of the angles of a convex quadrilateral is

$$360^\circ$$



Polygon Interior Angle Theorem (PIAT)

The sum of the measures of the angles of a convex polygon with n sides is

$$(n - 2) \times 180^\circ$$

Regular Polygon Interior Angle Theorem (RPIA)

The measure of each angle of a regular n -gon
is

$$\frac{(n - 2)180^\circ}{n}$$

Polygon Exterior Angles Theorem (PEAT)

The sum of the measures of the exterior angles, one at each vertex, of any convex polygon is

$$360^\circ$$

Right Angles Congruency Theorem

Any two right angles are congruent.

Quick Reference

Abbreviations (1/2)

SP	Supplement Postulate
VAT	Vertical Angle Theorem
ASPP	Angle Sum of a Point Postulate
ST	Supplement Theorem
CP	Complement Theorem
PCAC	Corresponding Angles Postulate
PAIC	Alternate Interior Angles Theorem
PAEC	Alternate Exterior Angles Theorem

Abbreviations (2/2)

PSSIAS	Same-Side Interior Angles Theorem
TIAT	Triangle Interior Angle Theorem
EAT	Exterior Angles Theorem
QIAT	Quadrilateral Interior Angle Theorem
PIAT	Polygon Interior Angle Theorem
RPIA	Regular Polygon Interior Angle Theorem
PEAT	Polygon Exterior Angles Theorem

Thank you!